

THE EFFECT THAT SIDE DOMINANCE HAS ON BARBELL POWER SYMMETRY DURING THE HANG POWER CLEAN

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ABSTRACT

Lake, JP, Lauder, MA, and Smith, NA. The effect that side dominance has on barbell power symmetry during the hang power clean. *J Strength Cond Res* 24(11): 3180–3185, 2010—The aim of this study was to examine whether ground reaction force (GRF) side differences were transmitted and related to bar end power output asymmetries during hang power clean (HPC) performance and whether progressive loading would intensify this effect. Differences between the dominant (D) and nondominant (ND) side average GRFs (AGRFs) of both feet and average bar end power outputs were recorded simultaneously from 15 recreationally trained male volunteers at 30, 60, and 90% 1RM using 2 force platforms and 3 high-speed digital cameras, quantifying side dominance from perceived handedness (left- or right-side dominance [LRSD]), GRF side dominance (force side dominance [FSD]), and bar end power output side dominance (barbell side dominance [BSD]). With the exception of the LRSD condition, differences between the D and ND side AGRFs were significant (FSD: 1.8–4.3%; BSD: 5.1–6.4%, $p < 0.05$). Bar end power output side differences were significant for all conditions (LRSD: 1.5–5.4%; FSD: 0.5–3.4%; BSD: 3.9–5.6%, $p < 0.05$). Progressive loading did not significantly affect GRF side differences or the FSD average bar power side differences. However, during the LRSD and BSD conditions, the 60 and 90% side average bar power side differences were >the 30% equivalents. Average GRF side differences were not related to bar end power output side differences. Because of the consistent side difference of 4–6% investigators and strength and conditioning practitioners should exercise caution when interpreting changes in bar end power output.

KEY WORDS asymmetry, strength imbalance, weightlifting, ground reaction force

INTRODUCTION

There has recently been an increased research focus on the symmetry of lower-body resistance exercise (1,8–11). Investigators have determined side dominance by studying independently measured ground reaction forces (GRFs) of both feet in accordance with either perceived handedness (1,11) or GRF side dominance (8,11).

Newton et al. (11) used both of these approaches to establish GRF side differences during relatively heavy back squat performance (80% 1RM). They found that although healthy individuals tended to favor a dominant side by around 6%, this did not correspond with perceived right-side dominance (right handed). These results were replicated in another study by Flanagan and Salem (1), who reported left side dominance in a predominantly right handed (17 of 18 subjects) individuals during back squat performance across a range of loading conditions (25–100% 3RM). In a case study of a well-trained power lifter, Lake et al. (8) reported GRF side differences of up to 57% that were not consistent with perceived side dominance.

This is an important area of research that has highlighted the mechanical adaptations that appear to occur as a consequence of the unilateral demands of sporting performance, leg length discrepancy, or past injury, and the potential implications that these factors may have for sporting longevity (1,11). However, little is known about whether ground kinetic side differences are transmitted to the barbell during resistance exercise performance and whether this could affect bar end symmetry.

Lake et al. (9) used 2 force platforms to independently measure left- and right-side GRFs, in addition to 3 high-speed digital cameras to independently measure left and right bar end power outputs during progressively loaded back squat performance (30, 60, and 90% 1RM). They found differences between the dominant (D) and nondominant (ND) side average GRFs (AGRFs, -6.2 to 20.7% , $p < 0.05$) and bar end power outputs (-1.4 to 3.5% , $p > 0.05$) that were not consistent with perceived dominance (9). Of more practical relevance, these findings suggested that relatively small changes in power output may occur because of bar end asymmetry and in turn should be interpreted with caution.

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Figure 1. The start position, the conclusion of the pull phase, and catch position of the hang power clean.

To date, side dominance has not been quantified during variations of the Olympic weight lifts like the hang power clean (HPC). During HPC performance, the bar is displaced at a considerably greater rate and over a larger range of motion that tends to occur independently of the body (2,3). This may exacerbate AGRF and bar end power output side differences and as such warrants research attention.

Therefore, the aim of this study was to establish AGRF and bar end power output side differences during HPC performance from perceived and measured side dominances and to establish whether progressive loading would intensify these differences. A secondary aim of this study was to establish whether AGRF side differences were related to differences in bar end power outputs.

METHODS

Experimental Approach to the Problem

Independently measured GRF and bar end power outputs were used to establish the side differences of 15 male volunteers during HPC performance. Side dominance was determined using perceived handedness

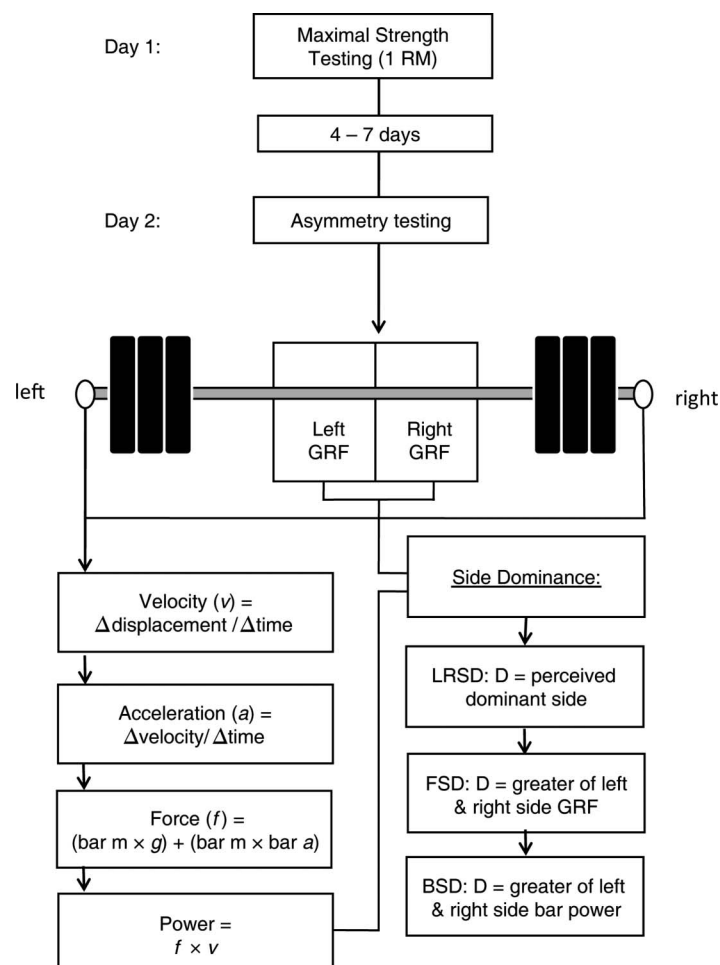


Figure 2. A schematic of the experimental time line, calculations, and the way in which side dominance was determined.

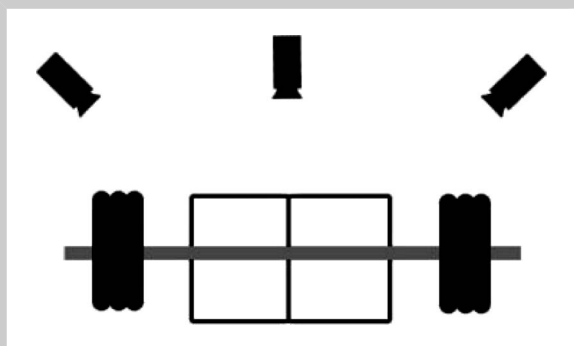


Figure 3. A schematic of the experimental setup that shows the position of the 3 cameras and 2 force platforms relative to the position of the bar during HPC performance.

(left- or right-side dominance [LRSD]), left and right GRF side dominance (force side dominance [FSD]) and left and right bar end power output side dominance (barbell side dominance [BSD]), where side dominance was defined as the highest left- or right-side AGRF or average bar end power (ABP). A loading spectrum of 30, 60, and 90% 1RM was used to establish the effect of progressive loading.

Subjects

Fifteen recreationally trained men (13) with a minimum of 1 year's HPC experience volunteered to participate in this study. Their mean ($\pm SD$) physical characteristics were age: 28.8 (± 8.9) years, mass: 78.2 (± 11.2) kg, height: 180 (± 0.03) cm, and HPC 1RM: 73 (± 15.5) kg. University of Chichester ethics approval was obtained before data collection, and all participants completed a health history questionnaire and provided written informed consent.

Procedures

All subjects participated in 2 testing sessions that were separated by approximately 7 days: During the first session, HPC 1RM was established using a procedure and performance technique that was similar to that outlined and used by Kawamori et al. ([7]; see Figure 1); during the second session asymmetry testing was performed (Figure 2).

During asymmetry testing, each participant performed 2 maximal effort single HPCs with 30, 60, and 90% 1RM with a minimum of 1 minute and a maximum of 3 minutes recovery between each lift (12). The 30, 60, and 90% 1RM loads were selected to encompass a light, moderate, and heavy spectrum of relative exercise intensity. Verbal encouragement was given during all performances (6,7).

Instrumentation

The vertical GRFs of HPC performance were recorded from both feet independently by 2 0.4 by 0.6-m Kistler 9851 force platforms (Alton, United Kingdom) at a sampling frequency of 500 Hz. The analog GRF signals were amplified by 2 type 9865E 8-channel charge amplifiers before they were digitally converted. Both force platforms were zeroed relative to the bar and participant system weight immediately before each trial.

Three digital cameras (Basler A602fc-2, Ahrensburg, Germany) were positioned on rigid tripods around and approximately 5 m from the center of the area of interest (Figure 3). Each camera filmed HPC performance at 100 Hz with a shutter speed of 1/1,000 seconds (4) after first recording a 17 point calibration frame (Peak Performance Technologies Inc., Englewood, CO, USA). The GRF and bar end kinematics were synchronized using a Vicon MX control unit (Peak Performance Technologies Inc.).

TABLE 1. Mean ($\pm SD$ and $\pm 95\%$ confidence interval) D and ND side concentric phase AGRF (N) and ABP (W).*

	AGRF 30%		AGRF 60%		AGRF 90%		ABP 30%		ABP 60%		ABP 90%	
	D	ND	D	ND	D	ND	D	ND	D	ND	D	ND
LRSD												
Mean	776.8	761.8	943.5	900.0	1,049.1	1,021.7	737.9	725.6	1,078.1	1,017.6	1,076.3	1,032.7
$\pm SD$	164.4	130.8	200.0	138.1	273.1	215.1	192.2	184.8	199.0	179.9	235.4	202.6
$\pm 95\%$ CI	74.7	59.5	90.9	62.8	124.1	97.7	87.3	84.0	90.4	81.8	107.0	92.1
FSD												
Mean	792.3	746.2	953.5	890.1	1,077.4	993.5	733.8	729.7	1,066.2	1,029.6	1,073.9	1,035.1
$\pm SD$	152.4	141.0	193.5	143.2	245.7	239.0	189.3	187.9	205.7	175.7	233.0	205.9
$\pm 95\%$ CI	69.3	64.1	87.9	65.1	111.6	108.6	86.0	85.4	93.5	79.9	105.9	93.6
BSD												
Mean	777.9	760.6	943.5	900.0	1,057.3	1,013.5	746.4	717.0	1,078.1	1,017.6	1,086.2	1,022.8
$\pm SD$	163.7	131.5	200.0	138.1	246.1	244.2	192.6	183.4	199.0	179.9	230.6	205.3
$\pm 95\%$ CI	74.4	59.8	90.9	62.8	111.8	111.0	87.5	83.3	90.4	81.8	104.8	93.3

*D = dominant side; ND = nondominant side; AGRF = average ground reaction force; ABP = average bar power; LRSD = perceived handedness; FSD = force side dominance; BSD = ABP side dominance; CI = 95% confidence interval.

Statistical Analyses

Retroreflective markers that were positioned on both ends of the bar were digitized at 100 Hz using Peak Motus 9.2 software from approximately 10 frames before the start of HPC performance to approximately 10 frames after its conclusion. After digitization, the raw coordinate data were smoothed using a low-pass filter with a cut-off frequency of 6 Hz following the recommendations of Winter (15) and differentiated with respect to time to obtain bar end velocity and acceleration (Figure 2).

Bar end power was calculated by multiplying bar end force by its velocity. Bar end force was calculated using the methods recently described and used by Hori et al. (5) where the mass of the bar was multiplied by its acceleration and added to the product of bar mass multiplied by gravitational acceleration (g):

$$\text{Force} = (\text{barbell mass} \times g) + (\text{barbell mass} \times \text{barbell acceleration}).$$

Measures of GRF and bar end power were then averaged across the duration of the concentric phase for further analysis. This approach has recently been used by Flannagan and Salem (1), who suggested that peak performance data may not accurately represent the behavior of parameters of interest over a selected period of time. The concentric phase was determined from the bar end velocity–time curve, with the transition from negative to positive velocity indicating the beginning of the concentric phase and peak instantaneous velocity the end of the concentric phase (3). Recent test–retest comparisons of both AGRF and ABP in lower-body resistance exercise demonstrated a high degree of both absolute (coefficient of variation: AGRF, 30% 1RM–1.3; 60% 1RM–0.9; 90% 1RM–1.2; ABP, 30% 1RM–6.5; 60% 1RM–6.5; 90% 1RM–8.3) and relative (intraclass correlation: AGRF, 30, 60, and 90% 1RM–0.99; ABP, 30% 1RM–0.95; 60% 1RM–0.94; 90% 1RM–0.91) reliability (9).

Side dominance was determined using 3 different methods: perceived handedness (LRSD) (1,11); left- and right-side concentric phase GRF dominance (FSD) (1,11); and left and right concentric phase bar end power output dominance (BSD). Differences between the left and right and D and ND side AGRF and ABP were then calculated as percentage differences using standard procedures.

Bar end power and GRF were presented as means ($\pm SD$ and 95% confidence intervals). A 2-way (side $\times$ load)

TABLE 2. Mean ($\pm SD$ and $\pm 95\%$ confidence interval) percentage differences between the D and ND side concentric phase AGRF and ABP.*

	AGRF 30%	AGRF 60%	AGRF 90%	ABP 30%	ABP 60%	ABP 90%
LRSD						
Mean	0.9	3.6	0.2	1.5	5.4	3.5
$\pm SD$	8.3	7.1	15.4	4.3	2.8	5.9
$\pm 95\%$ CI	3.8	3.2	7.0	2.0	1.3	2.7
FSD						
Mean	1.8	3.6	4.3	0.5	3.0	3.1
$\pm SD$	8.4	7.4	6.9	4.6	5.6	6.1
$\pm 95\%$ CI	3.8	3.4	3.1	2.1	2.6	2.8
BSD						
Mean	5.1	6.4	6.0	3.9	5.4	5.6
$\pm SD$	6.2	4.9	10.4	2.0	2.8	3.6
$\pm 95\%$ CI	2.8	2.2	4.7	0.9	1.3	1.7

*AGRF = average ground reaction force; ABP = average bar power; LRSD = perceived handedness; FSD = force side dominance; BSD = ABP side dominance; CI = 95% confidence interval

repeated-measures analysis of variance with Holm–Sidak post hoc comparisons was used to identify statistical differences between perceived and measured D and ND side AGRF and ABP across different load conditions. Pearson product–moment correlations were used to assess the relationship between the D and ND AGRF and ABP perceived and measured side differences. All statistical calculations were performed using SPSS version 16.0 for Windows (SPSS, Inc., Chicago, IL, USA) and an alpha value of $p \leq 0.05$ was used to determine statistical significance.

RESULTS

The mean $\pm SD$ and 95% confidence intervals are presented in Table 1 and the mean ($\pm SD$ and 95% confidence intervals) differences between the LRSD, FSD, and BSD dominant and nondominant side AGRF and ABP are presented in Table 2.

The results show that when side dominance was determined using perceived handedness (LRSD), the

TABLE 3. Pearsons product–moment correlations between the AGRF and ABP D and ND side differences.*

	30%	60%	90%
FSD	–0.23	–0.11	0.18
LRSD	–0.12	–0.25	0.46
BSD	–0.51	–0.25	0.24

*FSD = force side dominance; LRSD = left-right side dominance; BSD = bar power side dominance.

differences between the D and ND side AGRFs were not significant (0.2–3.6%; $p = 0.107$; $\eta = 0.175$; $\beta - 1 = 0.361$); however, D side ABP was significantly greater than the ND side ABP (1.5–5.4%; $p < 0.001$; $\eta = 0.785$; $\beta - 1 = 1.00$). Relative exercise intensity did not significantly affect the pattern or magnitude of the AGRF side differences ($p = 0.569$; $\eta = 0.039$; $\beta - 1 = 0.136$), but its effect on ABP side differences approached statistical significance ($p = 0.057$; $\eta = 0.185$; $\beta - 1 = 0.561$) with side differences that tended to be greater in the 60 and 90% 1RM condition compared to the 30% 1RM condition (Table 2).

Determining side dominance according to left- and right-side GRF dominance (FSD) resulted in significant differences between the D and ND side AGRFs (1.8–4.3%; $p < 0.001$; $\eta = 0.713$; $\beta - 1 = 1.00$) and ABP (0.5–3.4%; $p = 0.006$; $\eta = 0.428$; $\beta - 1 = 0.852$). However, relative exercise intensity did not significantly affect the pattern or magnitude of the AGRF ($p = 0.216$; $\eta = 0.104$; $\beta - 1 = 0.313$) or ABP ($p = 0.215$; $\eta = 0.104$; $\beta - 1 = 0.314$) D and ND side differences.

When side dominance was determined using bar end power output (BSD), the results showed that there were significant differences between the D and ND side AGRFs (5.1–6.4%; $p = 0.042$; $\eta = 0.264$; $\beta - 1 = 0.55$) and ABP (3.9–5.6%; $p < 0.001$; $\eta = 0.883$; $\beta - 1 = 1.00$). Relative exercise intensity did not significantly affect the pattern or magnitude of AGRF side dominance ($p = 0.535$; $\eta = 0.044$; $\beta - 1 = 0.146$) but significantly affected the ABP side differences ($p = 0.035$; $\eta = 0.213$; $\beta - 1 = 0.643$); the 60 and 90% 1RM side differences were significantly > the 30% 1RM side differences ($p < 0.0001$) (Table 2).

The relationships between AGRF and ABP are presented in Table 3. The results of the correlation analysis showed that there were no significant relationships between AGRF and ABP D and ND side differences. However, progressive loading appeared to alter the direction of the relationships from negative to positive (Table 3).

DISCUSSION

This study set out to establish AGRF and ABP side differences during HPC performance from perceived and measured side dominances and to establish whether progressive loading would intensify these differences. A secondary aim of this study was to establish whether AGRF side differences were related to differences in bar end power outputs.

In agreement with recent research (9,11), there were no differences between the perceived D and ND side AGRFs. Newton et al. (11) recently explained that when perceived side dominance does not correspond with measured side dominance, differences could be nullified. In the present study, all participants reported perceived right side dominance, but only 7 demonstrated this across all conditions; the remainder favored their left and right sides interchangeably, suggesting that side dominance should not be determined from perceived handedness.

Differences between the measured ground kinetic differences (FSD) were lower than those reported in recent research (9,11), remaining below 5%. Further, differences varied both within and between subjects indicating an individualized response to progressive loading. Only 6 demonstrated AGRF side differences that were consistent during the different loading conditions, with 6 demonstrating nonsignificant increases (3.3–14.8%) as load increased, and the remaining three, reductions (2.4–13.6%) in response to load increases.

These differences suggest that movement symmetry should be assessed as part of the strength and conditioning process and that assessment should use key resistance exercises across a range of loads.

The measured AGRF side differences were lower than those recently reported for back squat performance across the same loading conditions (1,9,11). Maximal back squat strength and in turn training loads tend to be considerably greater than their HPC equivalents. This may intensify ground kinetic side differences, although further research using a similar participant population would be needed to clarify this.

Another factor may be the technical demands of HPC performance, in that it requires a more coordinated and relatively symmetrical application of force to achieve the appropriate bar and body displacements. Winchester et al. (14) recently demonstrated that improvements in Olympic lifting technique can improve the efficiency of force application during power clean performance. This suggests that technical proficiency in the HPC is related to the magnitude of ground kinetic side differences and could be an area warranting further research.

Differences in bar end power output were not affected by the way in which side dominance was determined so that both perceived and measured side differences demonstrated statistical significance. However, the pattern of these differences varied, which may explain why AGRF and ABP differences were not related.

Data reported by Lake et al. (9) suggest that transmission of ground kinetic side differences to bar end power outputs is attenuated during back squat performance. For example, ground kinetic side differences of 21% resulted in bar end power output side differences that did not exceed 2.5% (9). In the present study, mean AGRF side differences of 6–8% resulted in ABP output side differences of between 4 and 6%, suggesting that HPC bar end symmetry is more susceptible to ground kinetic side differences than its back squat equivalent. This appears to be a consequence of the way in which HPC and back squat bar displacements differ, suggesting that the bar end power outputs from resistance exercises where the bar is displaced independently of the body should be interpreted with caution.

Therefore, the ABP output side difference across the different loading conditions was 5% with a 95% confidence interval of $\pm 0.9\%$. The applied relevance of this finding was

that in 95% of HPC performances across a range of loads practitioners and investigators should expect differences between bar end power outputs of between 4 and 6%.

These findings highlight the need to assess movement symmetry during the strength and conditioning process, while urging caution in the way in which differences between bar end power outputs are assessed. Practitioners should consider the findings of this study when selecting methods to assess differences in bar end power outputs.

Subsequent investigations should examine joint kinetic responses to ground kinetic side differences and the role that strength imbalances, whether a consequence of leg length discrepancy or past injury, and technical proficiency have in ground kinetic and bar end power output side differences.

PRACTICAL APPLICATIONS

Healthy individuals demonstrate significant ground kinetic side differences during HPC performance. Furthermore, power outputs obtained from both ends of the barbell differed by 4–6%. This suggests that relatively small changes in power output should be interpreted with caution unless a D side is identified and considered during power testing. Furthermore, movement symmetry should be assessed as part of the strength and conditioning process.

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